

# How topology maps to quantum state space

*State Layer (Hilbert Space) | Version 1.5*

ZP Companion | May 2026

This companion explains the ideas in plain language with diagrams and real-world examples. It is not the formal ontology — every claim here restates a result already proven in the corresponding technical document. Consult that document for the authoritative mathematics.

## Background: Vectors and Orthogonality

ZP-D works with objects called vectors. A vector is simply a quantity that has both a size and a direction — an arrow in space. You can add vectors (combine the arrows tip-to-tail) and scale them (stretch or shrink). A vector space is a collection of vectors where these operations always produce another vector in the collection.

The key relationship between vectors is orthogonality — which just means perpendicular. Two vectors are orthogonal when they share no common direction: pointing entirely independently of each other. In 2D, North and East are orthogonal. In 3D, you can have three mutually orthogonal directions. In higher-dimensional abstract spaces the same idea extends: orthogonal vectors are completely independent, with zero overlap.

Orthogonality is measured by the inner product  $\langle u, v \rangle$ : a number that captures how much two vectors "point in the same direction." When  $\langle u, v \rangle = 0$ , the vectors are orthogonal — they have nothing in common.

### Real-world example — Compass directions

North and East are orthogonal — perpendicular, sharing nothing. If you know how far north something is, that tells you nothing about how far east it is. ZP-D says: states that are clopen-separated in  $\mathbb{Q}_2$  (placed in different open-and-closed sets, with no continuous path between them) should be represented as orthogonal vectors in  $H$  (no shared component). Geometric independence matches clopen separation.

## What Is ZP-D Doing?

ZP-D builds a bridge between the  $p$ -adic topology of ZP-B and a Hilbert space  $H = \mathbb{C}^n$ . A Hilbert space is a vector space equipped with an inner product — the operation that measures orthogonality. The  $\mathbb{C}^n$  notation means the vectors have  $n$  complex-number components rather than  $n$  real-number components. (Complex numbers extend the real number line by adding an imaginary dimension; they are the natural language of quantum mechanics, but the geometric intuition of vectors and perpendicularity carries over directly.)

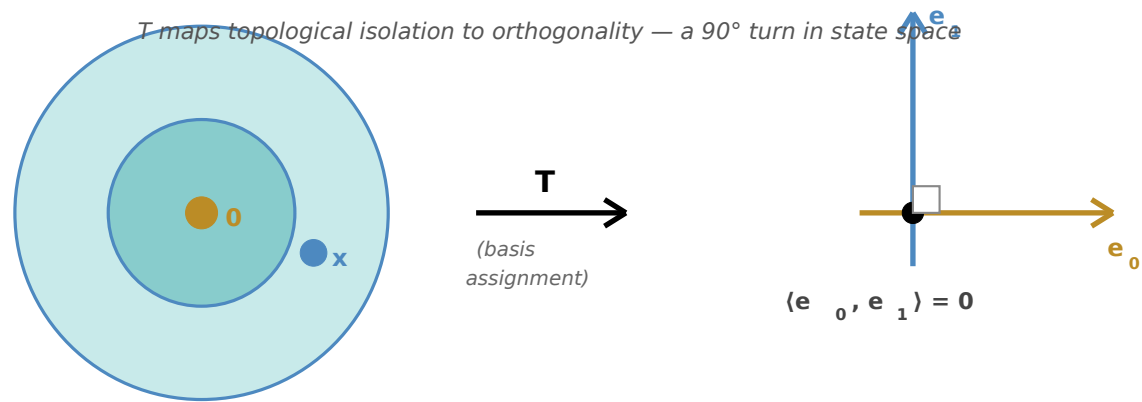
ZP-D constructs an explicit map  $T: \mathbb{Q}_2 \rightarrow H$  that carries the topological structure of  $\mathbb{Q}_2$  into the state geometry of  $H$ . The central insight: clopen separation in  $\mathbb{Q}_2$  corresponds to orthogonality in  $H$ . States with no continuous path between them are represented as perpendicular vectors.

ZP-D works with exactly two foundational states: the Null State ( $\perp$ , non-existence) and the first non-null state ( $\epsilon_0$ , existence). This is not a simplifying assumption — it is the irreducible minimum. The core question is whether any transition from  $\perp$  to a first state is possible, and for that question  $n=2$  is the smallest meaningful case. No claim in ZP-D requires  $n > 2$ . Binary is the logical ground floor, not a placeholder for a more general construction.

## The Transition Operator T

T is constructed by basis assignment: the clopen ball partition of  $\mathbb{Q}_2$  maps to an orthonormal basis of H. An orthonormal basis is a set of vectors that are mutually orthogonal (all inner products equal zero) and each have length exactly 1. Think of it as a coordinate system where every axis is perpendicular to every other and all axes are the same length.

The element 0 maps to  $e_0$ ; the first non-zero state  $\epsilon_0$  maps to  $e_1$ ; and so on. Because clopen balls are completely separated (ZP-B), their images under T are orthogonal in H.



T maps clopen separation (Def 1,  $\mathbb{Q}_2$ ) to orthogonality (right,  $H = \mathbb{C}^n$ ). 0 (amber) maps to  $e_0$ . Non-zero (blue) maps to  $e_1$ . Clopen separation becomes orthogonality:  $\langle e_0, e_1 \rangle = 0$ .   
~~isolation~~  $\rightarrow$  ~~orthogonality~~   
 isolation  $\rightarrow$  orthogonality

### Real-world example — Red and blue channels in an image

In digital images, the red and blue channels are orthogonal: changing one has no effect on the other. ZP-D maps topologically distinct states (no path between them) to independent dimensions in H (no overlap between vectors). Independence in topology becomes independence in geometry.

Remember: Image processing illustrates orthogonality, not the content of the framework. H is an abstract mathematical space; its elements become physical states only when the framework is instantiated with specific physical parameters.

### Key Result: T Exists and is Unique up to Rotation (T2, T3)

T can be explicitly constructed by basis assignment (T2). It is unique up to a rotation of the basis (T3) — meaning any two valid T operators differ only in which direction they label as "e<sub>0</sub>" vs. "e<sub>1</sub>" vs. "e<sub>2</sub>" etc. This is called unitary equivalence: like choosing different compass orientations for a map, the underlying geometry is the same. Two maps that agree on distances but point north differently are equivalent in all the ways that matter.

R3 — T respects the topology: A map between topological spaces can introduce discontinuities that were not present in the original space. R3 establishes that T does not do this — T is locally constant and continuous with respect to the p-adic topology of  $\mathbb{Q}_2$ . It does not invent jumps or boundaries that are absent from the source structure. The orthogonal geometry of H reflects exactly the topological structure of  $\mathbb{Q}_2$ , nothing more.

#### Key Result: The Snap Produces an Orthogonal Shift (T4)

The Binary Snap —  $0 \rightarrow \epsilon_0$  in  $\mathbb{Q}_2$  — maps to a shift from  $e_0$  to  $e_1$  in H such that  $\langle e_0, e_1 \rangle = 0$ . The Snap is a right-angle turn in state space.

#### Key Result: Every Genuine Transition Produces an Orthogonal Shift (T5-b)

The Snap is not a special case. For any state sequence, whenever two consecutive states are distinct — meaning the sequence actually moved — their T-images in H are orthogonal. Each genuine transition opens a new direction. The whole ascending chain, not just the first step, unfolds through orthogonal shifts. Lean: ZPD.t5\_strict\_orthogonal (axiom-free, via DP-1).

Design Commitment DP-1: The choice to represent topological isolation as orthogonality is a design commitment — the natural and consistent choice, stated explicitly. Other faithful representations exist in principle. This honesty about what is chosen versus derived is central to the framework.